1. Which of the following statements about encapsulation is true?
   - A) It hides data within a single class.
   - B) It exposes the internal implementation of a class.
   - C) It focuses only on reusability of code.
   - D) It makes all attributes of a class public.

2. What will the following code output?

```
class Example {
    String message = "OOP Concepts";
    void displayMessage() {
        System.out.println(message);
    }
}
public class Test {
    public static void main(String[] args) {
        Example e = new Example();
        e.message = "Encapsulation";
        e.displayMessage();
    }
}
```

   - A) OOP Concepts
   - B) Encapsulation
   - C) Compilation Error
   - D) Runtime Exception

3. Which of the following best describes abstraction?
   - A) Hiding implementation details and exposing only functionality.
   - B) Making all methods private.
   - C) Encapsulating data into a single class.
   - D) Exposing internal logic for debugging.

4. What is the output of the following code?

```
abstract class Animal {
    abstract void sound();
}
class Dog extends Animal {
    void sound() {
        System.out.println("Woof");
    }
}
public class Main {
    public static void main(String[] args) {
        Animal a = new Dog();
        a.sound();
```

```
    }
}
```

- A) Compilation error
- B) Woof
- C) Runtime exception
- D) Abstract method cannot be instantiated

5. Consider the following code. What will it output?

```
class Parent {
    void display() {
        System.out.println("Parent");
    }
}
class Child extends Parent {
    void display() {
        System.out.println("Child");
    }
}
public class Main {
    public static void main(String[] args) {
        Parent p = new Child();
        p.display();
    }
}
```

- A) Parent
- B) Child
- C) Compilation Error
- D) Runtime Exception

6. Which of the following is not true about method overloading?
- A) It is a compile-time polymorphism.
- B) It allows methods with the same name but different parameters.
- C) It supports inheritance.
- D) It cannot change the return type.

7. What will happen if an interface method is declared as `private` in Java?
- A) It will work as expected.
- B) Compilation error.
- C) The method will not be accessible to implementing classes.
- D) The method will be treated as default.

8. What does the following code print?

```
interface A {
    default void show() {
        System.out.println("Interface A");
    }
}
class B implements A {
    public void show() {
        System.out.println("Class B");
    }
}
public class Main {
    public static void main(String[] args) {
        A obj = new B();
        obj.show();
    }
}
```

- A) Interface A
- B) Class B
- C) Compilation error
- D) Runtime exception

9. Which keyword is used to rethrow an exception in Java?
- A) throw
- B) throws
- C) catch
- D) finally

10. What does the following code output?

```
public class Test {
   public static void main(String[] args) {
      try {
         int data = 10 / 0;
      } catch (ArithmeticException e) {
         System.out.println("Exception caught");
      } finally {
         System.out.println("Finally block executed");
      }
   }
}
```

- A) Exception caught
- B) Finally block executed
- C) Exception caught Finally block executed
- D) Runtime exception

11. What is the correct way to write data to a file in Java?
- A) Using Scanner class
- B) Using FileOutputStream
- C) Using Thread class
- D) Using Console class

12. What does the following code snippet achieve?

```
BufferedReader br = new BufferedReader(new FileReader("input.txt"));
String line;
while ((line = br.readLine()) != null) {
   System.out.println(line);
}
br.close();
```

- A) Writes data to the file
- B) Appends data to the file
- C) Reads data from the file and prints it
- D) Deletes the file

13. Which of the following operations is not supported by the String class in Java?
- A) substring()
- B) reverse()
- C) toUpperCase()
- D) concat()

14. What will be the output of the following code?

```
String s1 = "Java";
String s2 = new String("Java");
System.out.println(s1 == s2);
```

- A) true
- B) false
- C) Compilation error
- D) Runtime exception

15. Which collection should you use if you need a sorted set of unique elements?
- A) ArrayList
- B) HashSet
- C) TreeSet
- D) HashMap

16. What is the output of the following code?

```
List<Integer> list = new ArrayList<>();
list.add(1);
list.add(2);
list.add(3);
list.remove(1);
System.out.println(list);
```

- A) [1, 2, 3]
- B) [1, 3]
- C) [2, 3]
- D) [1, 2]

17. Which is not a valid state in the thread lifecycle?
- A) New
- B) Blocked
- C) Waiting
- D) Terminating

18. What is the purpose of the `join()` method in Java?
- A) To merge threads
- B) To terminate a thread
- C) To wait for a thread to finish execution
- D) To start a thread

19. Which SOLID principle emphasizes a class should have only one reason to change?
- A) Single Responsibility Principle
- B) Open-Closed Principle
- C) Interface Segregation Principle
- D) Dependency Inversion Principle

20. What does the following code output?

```
class Singleton {
    private static Singleton instance;
    private Singleton() {}
    public static Singleton getInstance() {
        if (instance == null) {
            instance = new Singleton();
        }
        return instance;
    }
}
public class Main {
    public static void main(String[] args) {
        Singleton s1 = Singleton.getInstance();
        Singleton s2 = Singleton.getInstance();
        System.out.println(s1 == s2);
    }
}
```

- A) true
- B) false
- C) Compilation error
- D) Runtime exception